EthoLive: how the model works

One design with two species files. A first look picks the path, then both follow the same steps from picture to behavior. What differs is what each one was trained to see.

EthoLive

First look: Identify Animal

Mouse or fly? How many? One: tracking only. Two: full model. Three or more: count only.

`two animals: one design, one file per species`

Mouse model 0.4

`INPUT`

Video of two mice, 30 fps

Trained on CalMS21: top view, two mice, one black, one white

`pictures`

`NETWORK 1: THE EYES, LEARNED`

Convolutional neural network (CNN)

① Resize the picture to 384 x 640, with a grey border.
② Slide small filters over it: edges first, then body shapes.
③ At every spot, predict a box, a confidence, and 7 body points.
④ Keep the surest black mouse and the surest white mouse.
⑤ If a mouse is missed for up to 3 pictures, keep its last points.

`Size: 2.55 million learned parameters.`

`body points (x, y)`

`SCOPESCORE STEPS, FIXED CODE, NOT LEARNED`

ScopeScore: 9 methods make 120 numbers per frame

Quality flags	mark points that are missing, jump, or are unsure
Swap fix	nose is the end by the ears, tail base by the hips
One-Euro filter	removes jitter, using only past frames
Per-point smoothing	nose quick, tail base steady, the rest medium
Motion measures	speed, turning, pauses, distance, approach
Body axis	one straight line: length, width, where the other is
Spine center	the midpoint along the spine
Spine bend	straight or curled, from chord / spine length
Edge distance	how close the body is to the picture edge

`Adapted from tracking surgical tools: nose = tip, tail = start.`

`120 numbers per frame`

`NETWORK 2: THE BRAIN, LEARNED`

Small temporal convolutional network (TCN)

Frame branch
one linear layer: this frame only

Memory branch
5 dilated causal conv blocks, width 64
sees the past 63 frames (2.1 s)

+ → softmax

91,688 parameters per copy. 3 copies average their votes.
No lookahead: mouse behaviors last long enough.

`probabilities`

`OUTPUT, EVERY FRAME`

Attack, investigation, mount, or other

One probability for each, plus reliable-tracking flag.

Held-out test macro F1: 0.847

77 new videos: 0.773. About 1.2 false attack alerts per hour.

Fly model 0.10

`INPUT`

Video of two male flies, 30 fps

Trained on Fly vs. Fly: top view, Aggression and Boy meets boy

`pictures`

`NETWORK 1: THE EYES, LEARNED`

Convolutional neural network (CNN)

① Prepare the picture: 160 x 160, or larger so flies match training.
② Slide small filters over it: edges first, then body shapes.
③ At every spot, predict a box, a confidence, and 5 body points.
④ Keep the two surest detections that do not overlap.
⑤ Match them to last positions, so fly a stays fly a.

`Size: 2.49 million learned parameters.`

`body points (x, y)`

`SCOPESCORE STEPS, FIXED CODE, NOT LEARNED`

ScopeScore: 8 methods make 94 numbers per frame

Quality flags	mark points that are missing or are unsure
Swap fix	head is the end away from the wings
Per-point smoothing	head quick, abdomen steady, the rest medium
Motion measures	One-Euro filter, then speed, turning, distance
Body axis	head to abdomen: length, other fly in front or beside
Wing angles	left, right, spread, asymmetry, length
Facing	how straight each fly points at the other
Edge distance	how close the thorax is to the chamber edge

`First, points are rescaled to the training fly size.`

`94 numbers per frame`

`NETWORK 2: THE BRAIN, LEARNED`

Small temporal convolutional network (TCN)

Frame branch
one linear layer: this frame only

Memory branch
5 dilated causal conv blocks, width 64
sees the past 63 frames (2.1 s)

+ → softmax

90,080 parameters per copy. 3 copies average their votes.
Lookahead: sees 6 frames past the labelled frame (0.2 s).

`probabilities`

`OUTPUT, EVERY FRAME`

Lunge, wing threat, tussle, hold, or other

One probability for each, plus reliable-tracking flag.
Separate courtship model: wing extension, circling, attempts, copulation.

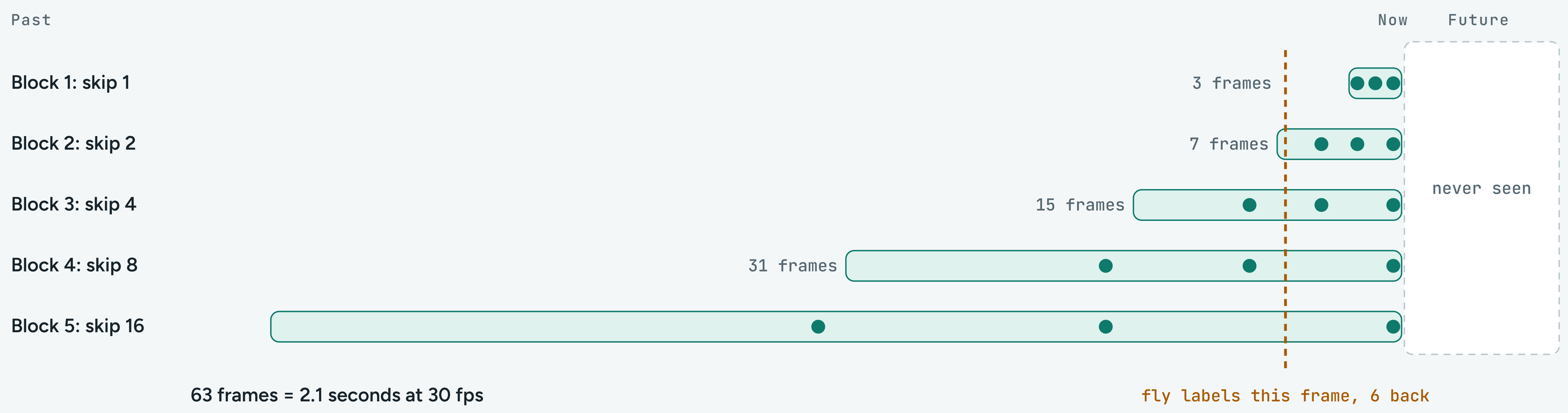
Validation macro F1, training camera: 0.66

No clean test yet. Second camera: lunge 0.50. Courtship: validation only.

Each model is two neural networks with fixed code between them. The first turns pictures into body points. The second turns body points over time into behavior.

How the memory branch sees 2.1 seconds

Each block looks at 3 frames, but later blocks skip further apart. Five blocks reach 63 frames with very few parameters. It only looks backward, so it can run live.



Dots are the 3 frames a block reads. Bars show how far back the network can see after that block, drawn to scale. For flies, the frame being labelled sits 6 frames behind "now", so the model sees how a lunge ends.